

Telecommunications Analytics Project

Telco Customer Churn & Sentiment Analysis

A comprehensive data-driven analysis examining customer churn patterns and sentiment indicators within the telecommunications industry. This project combines advanced analytics with customer feedback to identify critical retention opportunities and revenue protection strategies.



POWER BI



PYTHON

NLP

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Business Problem, Data & Analytical Approach

□ Business Problem

- Telecom companies face **high customer churn**, leading to **significant revenue loss**
- Traditional reports explain *who churned*, but not *why customers churn*.
- Business needs **early churn signals** to retain **high-value customers**

□ Analytical Approach

- **Data cleaning & feature engineering** (tenure groups, service count, value flags)
- **Sentiment analysis (NLP)** on customer feedback to detect emotional signals
- Integrated behavioral + sentiment insights in **Power BI dashboards**
- Focus on **churn risk, revenue impact, and retention opportunities**

📊 Data Used

- **Customer Demographics:** gender, senior citizen, partner, dependents
- **Service Details:** internet type, contract, payment method, service count
- **Financial Metrics:** monthly charges, total charges, tenure
- **Customer Feedback (Text):** used for **sentiment analysis**

🔧 Tools & Technologies

POWER BI

PYTHON

NLP

Machine Learning

Executive Insights: Churn Impact Overview

26.5%

Overall Churn Rate

Significant portion of customer base at risk

42%

Month-to-Month Risk

Highest churn among flexible contracts

89%

Two-Year Stability

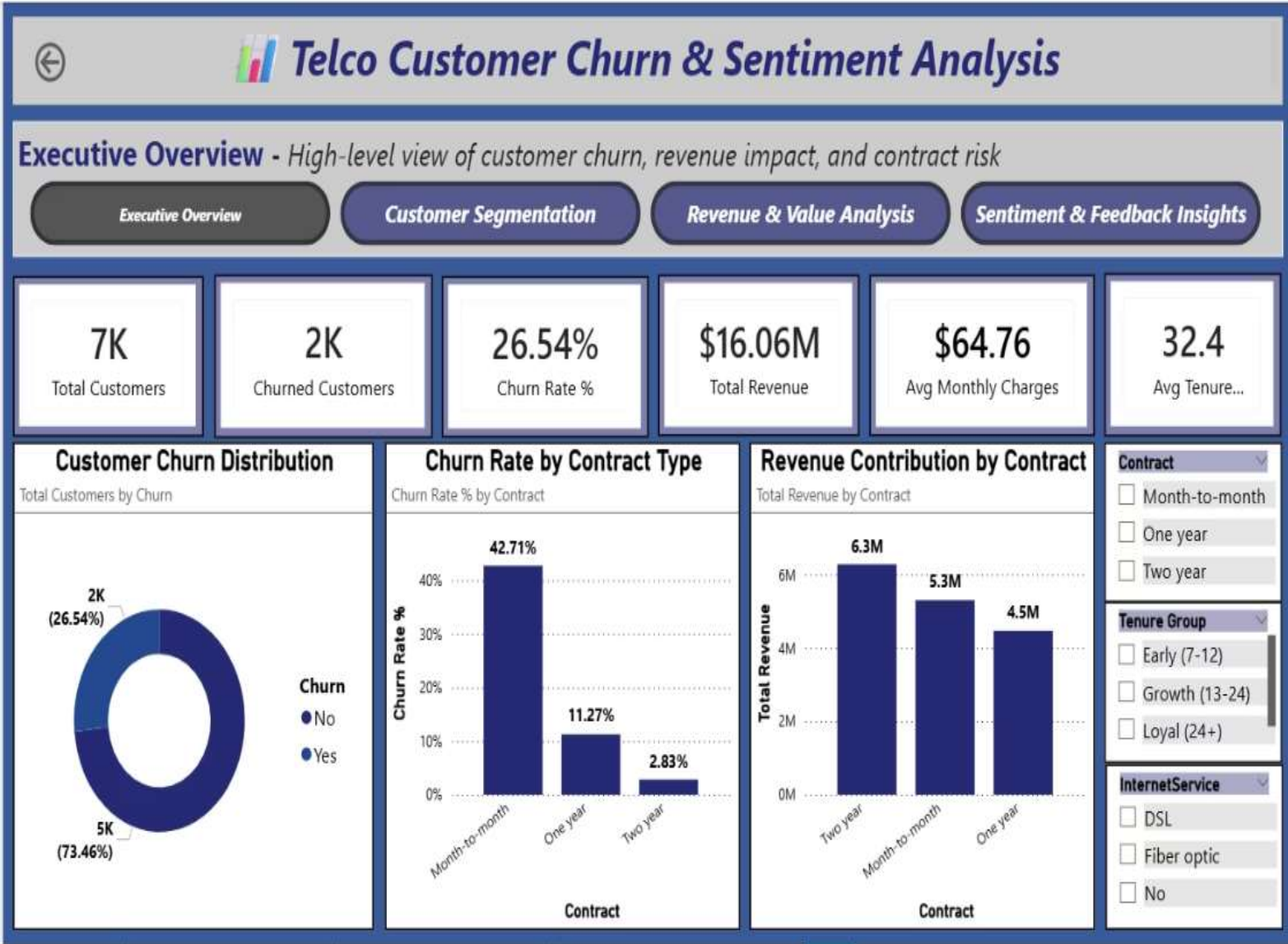
Long-term contracts drive retention

Key Findings from Executive Analysis

- The overall churn rate is **26.5%**, indicating a clear risk to revenue stability. Contract type plays a key role in customer retention.
- **Month-to-month customers churn the most**, creating higher revenue volatility, while **two-year contracts show strong customer loyalty** and stable revenue.
- Churn has a **high financial impact**, as customers who leave often contribute significant revenue, making targeted retention efforts essential.

Executive Takeaway

- Improving contract strategy and targeting high-risk customers can significantly reduce churn and protect long-term revenue.



Customer Segmentation: Identifying High-Risk Profiles

1

Early-Stage Customers

New customers in their first **0–6 months** show the **highest churn risk** during the first few months. This early onboarding phase is critical and requires stronger engagement and support to build long-term loyalty.

2

Fiber Optic Subscribers

Despite being a premium service, **fiber optic customers experience higher churn**. This suggests potential gaps in service expectations, pricing perception, or competitive alternatives.

3

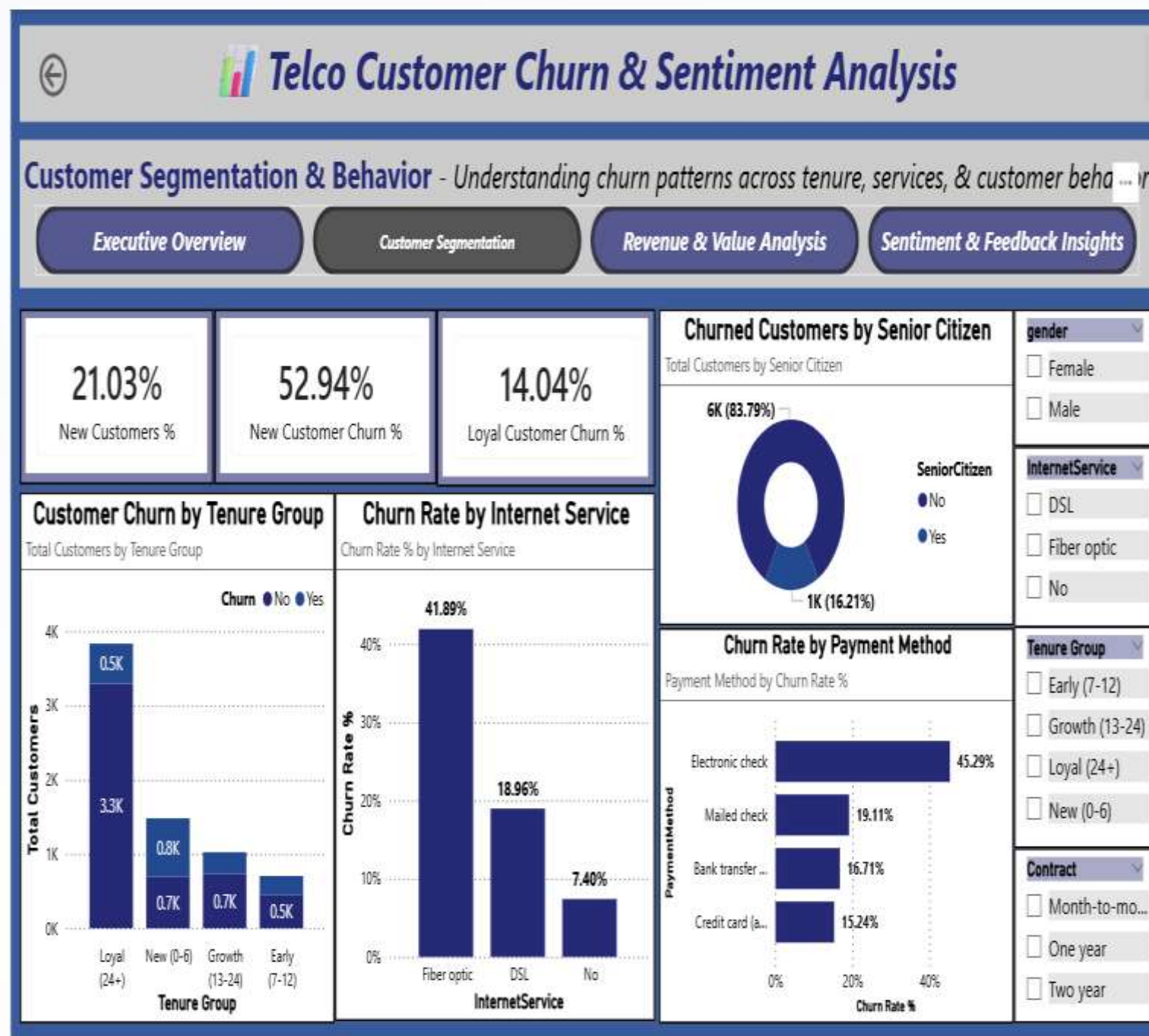
Electronic Check Users

Customers using **electronic check payments** exhibit elevated churn risk. Payment friction or billing experience may contribute to dissatisfaction and eventual churn.

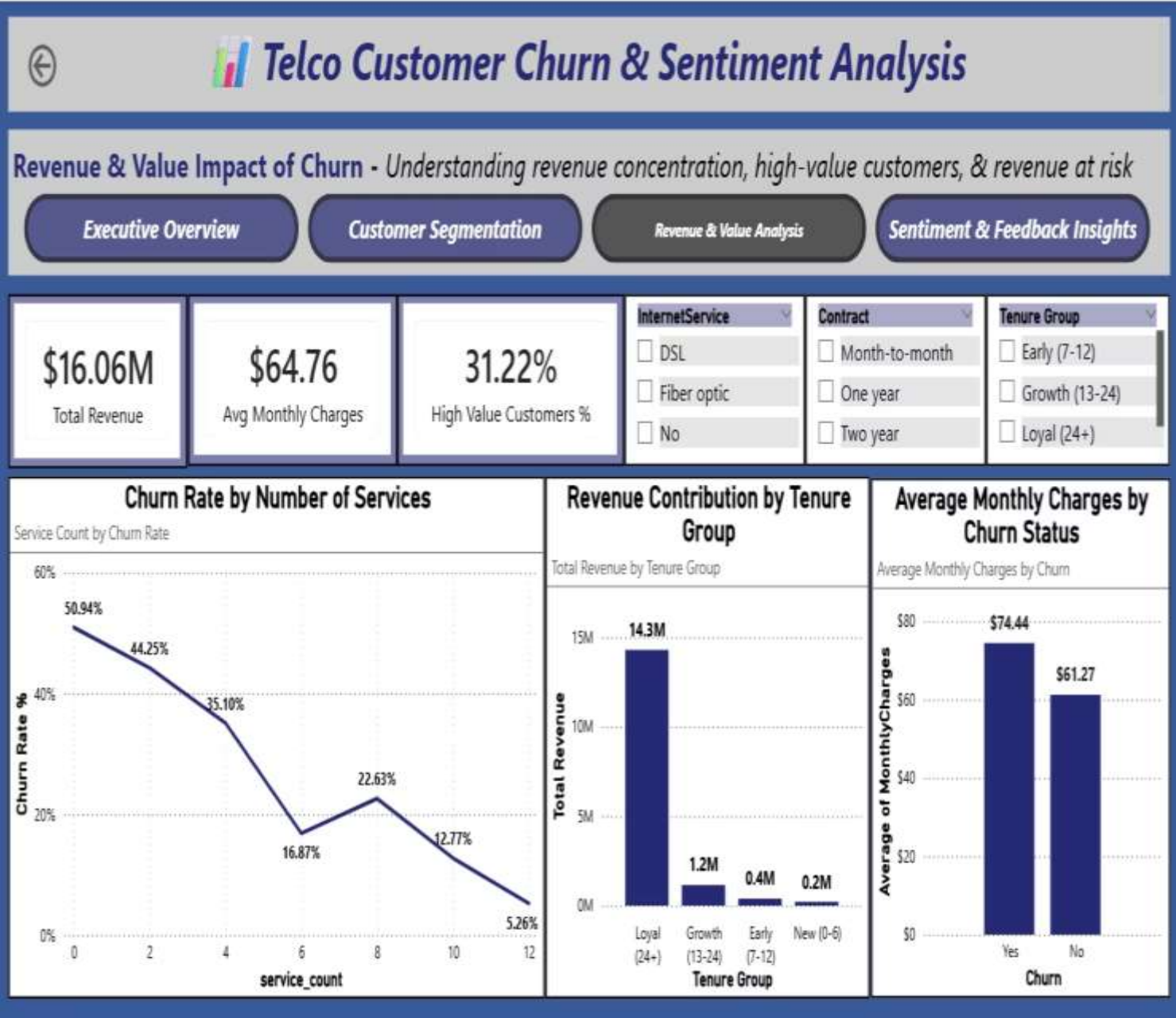
4

Senior Citizen Segment

Although a smaller customer group, **senior citizens show distinct churn patterns**. This segment benefits from targeted retention strategies tailored to their specific needs.



Revenue Impact: Understanding Value Distribution



Strategic Value Insights

- High-value customers form a smaller share (~31%) of the customer base but contribute a disproportionately large portion of total revenue, making them critical for revenue protection.
- Loyal customers are the backbone of stable revenue, with long-term relationships delivering predictable cash flow and higher lifetime value.
- Month-to-month churn causes significant revenue loss, highlighting the financial risk of short-term contracts.



Revenue from Loyal Customers
Majority revenue contribution from stable base



High-Value Customer Share
Critical segment for revenue protection



Revenue Loss from M2M
Month-to-month churn impact

Key Revenue Takeaways

- Retaining high-value and long-tenure customers has the **greatest impact on revenue stability**.
- Customers using **multiple services churn less**, indicating strong cross-sell and bundling opportunities.
- Revenue-focused retention strategies should prioritize **contract upgrades and service expansion**.



Business Intelligence Highlight: Customers with multiple services show lower churn rates, suggesting that bundling increases customer stickiness and reduces switching behavior.

Sentiment Analysis: The Early Warning System

Customer feedback reveals early signs of dissatisfaction that often **appear before churn happens**. By analyzing sentiment, we can identify at-risk customers and act proactively.



Negative Sentiment = High Churn Risk

Customers expressing negative sentiment show **significantly higher churn rates**, making sentiment a strong early warning signal.



High Charges + Negative Sentiment = Critical Risk

Many unhappy customers are also **high-paying**, increasing the financial impact if they churn.

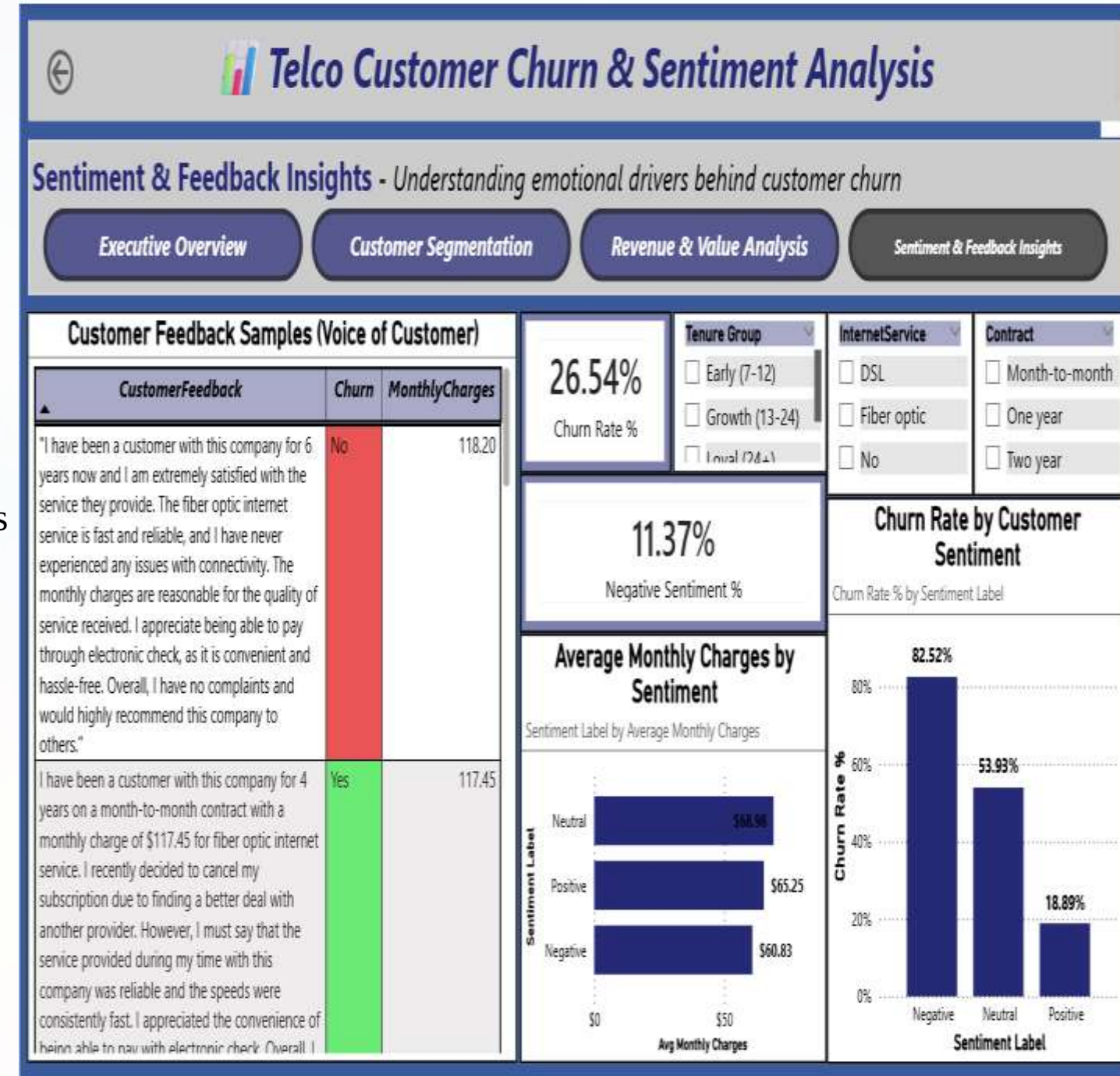


Customer Feedback Validation

Text feedback clearly highlights service issues and dissatisfaction, validating the churn patterns seen in quantitative metrics.

Strategic Advantage

Sentiment analysis enables **early warning mechanism**, allowing businesses to address customer issues before churn occurs—shifting retention strategies from reactive to proactive.



Business Recommendations: Strategic Action Framework

Based on comprehensive data analysis and sentiment insights, we propose a focused retention strategy addressing the most critical churn drivers and value protection opportunities.

1

Enhanced Onboarding Programme

First 6 months (highest churn risk)

- Structured onboarding for new customers
- Proactive check-ins during early tenure
- Educational guidance to improve service usage

✦ *Goal:* Reduce early-stage churn and build long-term loyalty.

2

Contract Migration Strategy

Month-to-month customers

- Incentives to move customers to yearly / two-year plans
- Value-based discounts for long-term commitments
- Simplified upgrade options

✦ *Goal:* Improve retention and stabilize revenue streams.

3

Sentiment-Driven Interventions

Customers showing negative sentiment

- Monitor sentiment from customer feedback
- Early alerts for dissatisfaction signals
- Priority support for at-risk customers

✦ *Goal:* Prevent churn before customers decide to leave.

4

High-Value Customer Focus

Top revenue-contributing customers (~31%)

- Dedicated support and faster resolution
- Loyalty benefits for premium customers
- Proactive relationship management

✦ *Goal:* Protect high-revenue customers and reduce financial loss.

Thank You

Driving Customer Retention Through Data Intelligence

Project Contact

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ML | NLP

Key Takeaway

Combining **sentiment analysis** with **customer behavior data** enables early churn detection and supports **proactive retention strategies**, helping protect revenue and strengthen long-term customer relationships.

🔗 QUESTIONS WELCOME

✉ CONTACT FOR DETAILS

